INCH-POUND

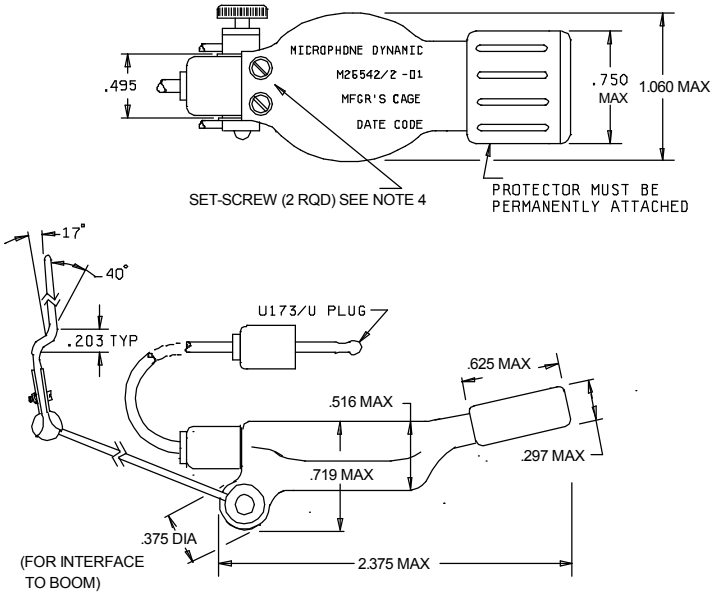
MIL-PRF-26542/2D
30 May 1997
SUPERSEDING
MIL-M-26542/2C
29 March 1985

PERFORMANCE SPECIFICATION SHEET

MICROPHONE AND MICROPHONE ASSEMBLIES, M87/AIC, M26542/2-01,M26542/2-02, M26542/2-03, AND M26542/2-04

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.



NOTES:

1. Quantity and configuration of sound ports optional.
2. Marking in accordance with part number supplied. Placement on surface shown, optional.
3. Dimensions are in inches. Tolerance is ± 0.015, unless otherwise specified.
4. Screws shall hold the element securely, shall be either slotted or allen type, and shall not protrude above the surface of the element.
5. Angular requirements of the boom shall be met to provide interface to headset, and for adjustability.

FIGURE 1. Microphone assembly

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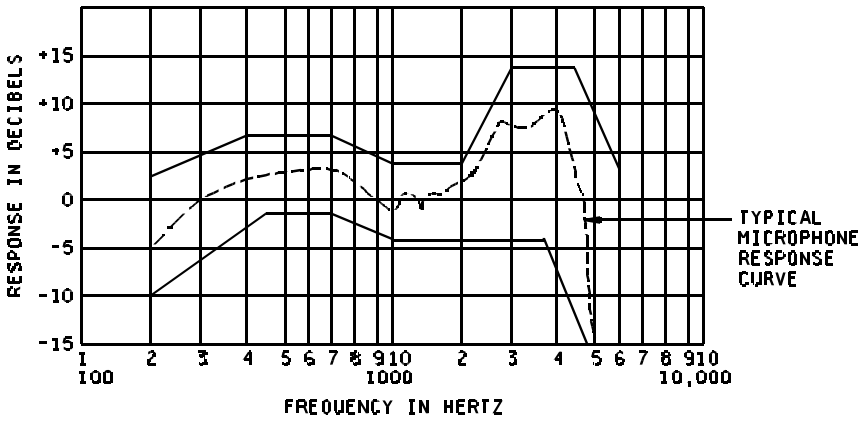


FIGURE 2. Frequency response at sea level

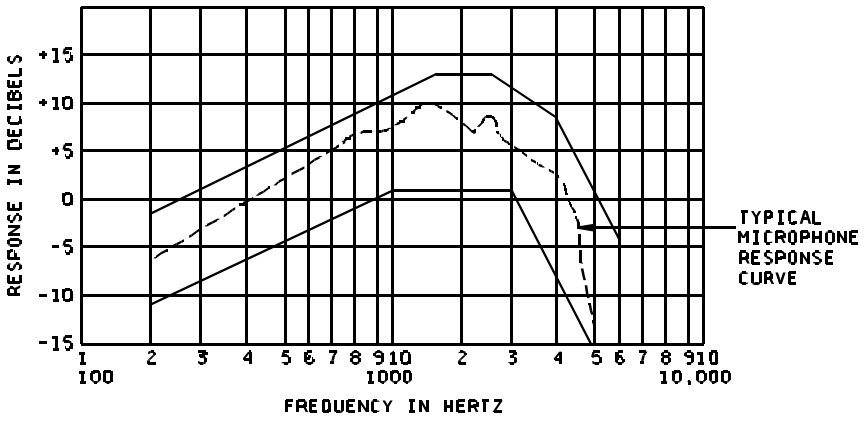


FIGURE 3. Frequency response at 25,000 feet

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REQUIREMENTS:

Component parts:

Microphone element: Shall be in accordance with figure 1.

Boom:

Dimensions:	Dimensions of the boom shall be in accordance with USAF Drawing 67B1854, for interchangeability with the next-higher-assembly headset-microphone.
Color:	Boom shall be the same color as the microphone element.
Finish:	Shall be in accordance with MIL-PRF-26542, paragraph 3.5.28.
Operation:	Shall be in accordance with MIL-PRF-26542, paragraph 3.5.29.
Material:	Shall be constructed from a high-strength, corrosion-resistant metal, meeting or exceeding the environmental and durability requirements of MIL-PRF-26542.

Cable assembly: Shall be in accordance with the specified part in order to provide interface to oxygen gear, for interchangeability, and for environmental performance. The microphone element shall provide a complete electrical and mechanical interface with the cable assembly. Cable assemblies per part numbers are listed in table I.

Plug assembly: U-173/U, in accordance with USAF Drawing 57B12662, or electrically and mechanically compatible part.

Weight: 45 grams, maximum.

Performance:

Sensitivity at ground level: -34.0 dB - 40.98 dB (re 1 mV) or 50.4 µV - 112.0 µV with a SPL input of 28 dynes/cm^2 at 1 kHz, when tested with themicrophone sound port .187 inches from, and coaxial with, the opening of the artificial voice.

Sensitivity at altitude: Sensitivity shall be within 3 dB relative to the ground level sensitivity when tested at 25,000 feet.

Frequency response at ground level: Shall be as shown in figure 2, when tested with the microphone port .187 inches from, and coaxial with, the artificial voice.

Frequency response at altitude: Shall be as shown in figure 3, with the microphone port .187 inches from, and coaxial with, the artificial voice.

Impedance: 4.0 ohms to 6.0 ohms.

Resistive load: 5.0 ohms.

Intended use: This is a noise-canceling dynamic microphone designed for use on a headband type headset at low altitudes or for use in a pressure type oxygen helmet, and at altitudes where the use of an oxygen helmet is required. It is not waterproof. The microphone is intended for use as a part of intercommunication set AN/AIC-10 ( ) to provide communication under the noise conditions encountered in military aircraft.

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The microphone assembly shall be tested in accordance with the tests listed in table II.

Marking Microphone assemblies shall be marked as shown on figure 1 and shall additionally be identified with a heat sealed band on the cord assembly showing the words "microphone assembly" and the microphone assembly part or identifying number (PIN), when supplied as an assembly.

TABLE I. Part number designations

Part or Identifying Number (PIN)	Characteristics
M-87/AIC	Supplied with microphone element, boom, and cable assembly M22442/36-1 (6.25 in).
M26542/2-01	Supplied with microphone element, boom, and cable assembly M22442/36-3 (13.00 in).
M26542/2-02	Supplied with microphone element, boom, and cable assembly M22442/36-4 (16.25 in).
M26542/2-03	Supplied with microphone element, boom, and cable assembly M22442/36-6 (23.00 in).
M26542/2-04	Supplied with microphone element only.

TABLE II. Parameter applicability

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1, 3.4, 3.5	4.5.1	X	X		
Sensitivity of ground level	3.5.1	4.5.2.2	X	X		
Sensitivity at altitude	3.5.2	4.5.2.3	X			
Frequency response at ground level	3.5.3	4.5.2.4	X	X		
Frequency response at altitude	3.5.4	4.5.2.5	X			
Impedance	3.5.5	4.5.3	X	X		
Noise cancellation characteristics	3.5.7.1	4.5.5.1	X			
Effect of external magnetic field	3.5.8	4.5.6	X			
Stray magnetic field	3.5.9	4.5.7	X			
Linearity	3.5.10	4.5.8	X			
Talk-out	3.5.11	4.5.9	X	X		
Dielectric withstanding voltage	3.5.12	4.5.10	X			
Signal-to-noise	3.5.7	4.5.5	X		X	
Distortion	3.5.6	4.5.4	X		X	
Interchangeability	3.5.13	4.5.11	X		X	

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TABLE II. Parameter applicability - continued

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group II						
Thermal shock	3.5.14	4.5.12	X			X
Humidity	3.5.15	4.5.13	X			X
Drop	3.5.16	4.5.14	X			X
Pressure equalization	3.5.17	4.5.15	X			X
Explosive decompression	3.5.18	4.5.16	X			X
Salt fog	3.5.19	4.5.17	X			X
Group III						
Vibration	3.5.20	4.5.18	X			X
Bounce	3.5.21	4.5.19	X			X
Altitude	3.5.22	4.5.20	X			X
Moisture barrier seal	3.5.23	4.5.21	X			X
Immersion	3.5.24	4.5.22	N/A			
Group IV						
Fungus	3.5.25	4.5.23	X			
Group V						
Gun blast	3.5.26	4.5.24	N/A			
Boom finish	3.5.28	4.5.26			X	
Boom operating force	3.5.29	4.5.27			X	

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME
Navy - AS, OS
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

(Project 5965-0238-02)

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